

A Complete Guide to Getting Started with Arduino

Building DIY projects with Arduino is super easy and a lot of fun. However, it may not always be as straightforward for beginners with absolutely no background knowledge in hardware or programming. In this article, we cover everything you need to know for getting started with your first Arduino project. That includes an overview of the Arduino board, the IDE, the programming language and a simple project to wet your beak.

Arduino is an open source project developed for students, hobbyists, artists and programmers who are interested in combining hardware and software to make DIY (Do It Yourself) devices and utilities. It was created in 2005 by Massimo Banzi and his colleagues and students at the Interaction Design Institute Ivrea, Italy. The primary purpose behind the creation of the Arduino project was to provide enthusiasts and professionals with a platform consisting of a programming language, an IDE, and a printed circuit board to create simple and effective DIY projects. The project quickly gained steam and by 2011, an estimated

300,000 Arduino boards were in circulation globally. By 2013, that number was 700,000.

Today, the Arduino project is a mammoth with 100+ devices consisting of various types of boards, shields, carriers, kits and other accessories. All the information regarding the Arduino project is available at its official website arduino.cc

So what makes Arduino so attractive? Go to YouTube and search ‘Cool Arduino Projects’ and the answer will become glaringly obvious. Arduino offers developers the freedom and autonomy to get creative with their circuit design, and write innovative code to make DIY devices and utilities

that not only help make life better but are also artistic and innovative in their approach.

Consider life before Arduino. If you wanted to build a light that turns on automatically, you would have to design the entire circuit diagram yourself. Solder it all together on a board. Write low level machine code yourself and create your own mechanism to power the whole thing up.

With Arduino, you can do all that in 5 minutes. Just connect a few pins here and there using an Arduino and a breadboard. Write the code in a language that is as simple as C and run the code with the click of a button. That’s it! Arduino massively simplifies your life as an engineer. What’s more, it is designed keeping students in mind, which means it is cheap to buy.

The anatomy of an Arduino device

Before we get to the code part, it is necessary to have a basic understanding of the Arduino board. Every Arduino board is differently constructed based on the purpose it serves. For this article, we shall be using the beginner-friendly and most popular Arduino board, i.e., the Arduino Uno. Figure 1 shows the anatomy of an Arduino Uno.

There are four major components of an Arduino Uno board — digital pins, analog pins, USB connector, and battery power.

Digital pins: As shown in Figure 1, there are 14 digital pins in the Arduino

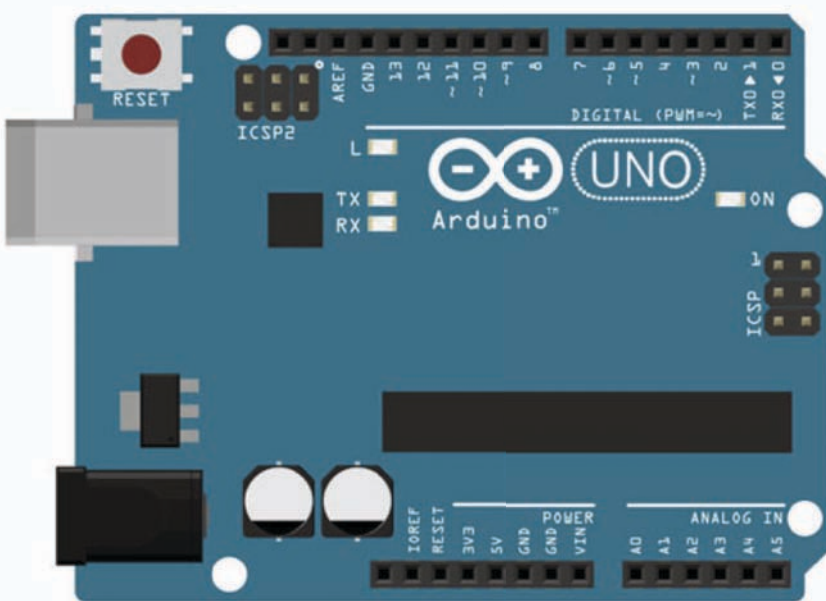


Figure 1: Arduino Uno